Recurrence relations

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First-order linear recurrences

For $a_n = pa_{n-1} + q$, p≠1 gives fixed point $a^* = q/(1-p)$. Subtracting it yields $b_n = pb_{n-1}$, hence

$$a_n = p^{n-n_0} a_{n_0} + q\frac{1-p^{n-n_0}}{1-p}.$$

For p=1, $a_n = a_{n_0} + (n-n_0)q$. Specify the initial index. Integer powers use $p^0=1$, including the empty-product convention $0^0=1$, without asserting a two-variable limit.

Unrolling for $n>n_0$ gives $a_n = p^2 a_{n-2} + pq + q = \cdots = p^{n-n_0}a_{n_0} + q(1 + p + \cdots + p^{n-n_0-1})$. The geometric sum gives the first formula for p≠1. For p=1 there are $n-n_0$ additions of q; p=0 gives $a_n=q$ after the initial index. Check the initial term, next term, and original recurrence. The fixed-point substitution is a convenient homogenization of the same unrolling proof.

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General forcing terms

For $a_n = pa_{n-1} + f(n)$, p≠0 allows the integrating factor

$$p^{-n}a_n - p^{-(n-1)}a_{n-1} = p^{-n}f(n).$$

Telescoping gives

$$a_n = p^{n-n_0}a_{n_0} + \sum_{k=n_0+1}^{n} p^{n-k}f(k).$$

The final formula no longer divides by p, so remains defined for p=0,1 and can be verified independently by substitution. Only the integrating-factor derivation needed p≠0. Negative p requires retaining parity signs in integer powers, rather than assuming positive-term geometric intuition.

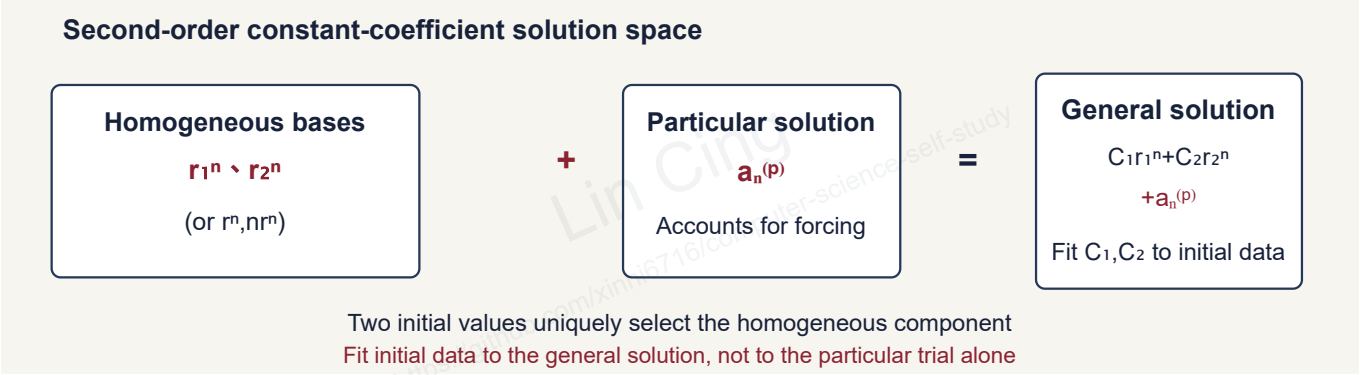
Variable coefficients and applications

For a_0=1 and a_n=na_{n-1}, unrolling gives a_n=n(n−1)…1a_0=n!. With a_0=0 every term is instead zero, showing why initial data are part of a recurrence specification.

Worst-case bubble-sort comparisons satisfy h_1=0,h_n=h_{n-1}+n−1: the first pass uses n−1 comparisons and leaves n−1 elements. Thus $h_n = \sum_{j=1}^{n-1} j = n(n-1)/2 = \Theta(n^2)$. This counts comparisons, not identical elapsed time on every machine; memory hierarchy and branch prediction still matter. Variable coefficients do not directly admit constant-coefficient characteristic roots; try products, telescoping, or substitutions and verify the actual per-round states.

Distinct characteristic roots

For $a_n + Aa_{n-1} + Ba_{n-2} = 0$, n≥2, trial r^n gives characteristic polynomial r^2+Ar+B. Distinct roots give $a_n = C_1 r_1^n + C_2 r_2^n$, with two initial conditions fixing constants. Check zero roots and the index domain separately.



Substitution gives $r^{n-2}(r^2 + Ar + B) = 0$, so a nonzero root supplies a solution. For nonzero distinct roots,

Distinct characteristic roots (continued)

$$\det \begin{pmatrix} r_1^n & r_2^n \\ r_1^{n+1} & r_2^{n+1} \end{pmatrix} = r_1^n r_2^n (r_2 - r_1) \neq 0.$$

They are independent. Every solution is uniquely determined by a_0,a_1, so the solution space has dimension two. The initial equations $C_1+C_2=a_0$,$C_1r_1+C_2r_2=a_1$ have determinant $r_2-r_1\neq 0$ and uniquely fit all solutions. Normalize the recurrence's signs before forming its polynomial.

Repeated, higher-order, and complex roots

A repeated nonzero root ρ gives $a_n = (C_1 + C_2 n)\rho^n$. To prove the second basis, use A=−2ρ,B=ρ²:

$$n\rho^n + A(n-1)\rho^{n-1} + B(n-2)\rho^{n-2} = \rho^{n-2}[n\rho^2 - 2\rho^2(n-1) + \rho^2(n-2)] = 0.$$

The initial-value matrix $\begin{pmatrix} 1 & 0 \\ \rho & \rho \end{pmatrix}$ has determinant ρ≠0, proving independence and completeness. If ρ=0, the recurrence only forces a_n=0 for n≥2, leaving a_0,a_1 arbitrary. The pair 0^n,$n0^n$ is not a basis.

For a nonzero root r of multiplicity m in an order-k recurrence, use r^n,nr^n,…,$n^{m-1}r^n$. Handle each root's multiplicity separately; the total basis count must equal k. For zero multiplicity m_0 use prefix sequences δ_0,…,δ_{m_0-1}, where $\delta_j(n)$=1 iff n=j. Conjugate complex roots can be paired into real trigonometric solutions. Fibonacci's roots are irrational real numbers, not nonreal conjugates.

General proof for higher-order solutions

Write the forward recurrence as P(E)a=0, with Ea_n=a_{n+1}, monic degree-k P, and n≥0. The first k arbitrary values determine the rest uniquely, so dimension is k. For nonzero r and polynomial q,

$$(E - r)(r^n q(n)) = r^{n+1}\Delta q(n).$$

The difference operator lowers positive degree by one and kills constants. Thus multiplicity m makes P(E) annihilate $r^n n^j$ for j<m. For another root λ≠r,

$$(E - \lambda)(r^n q(n)) = r^n[(r - \lambda)q(n) + r\Delta q(n)].$$

On polynomials of degree below m this map is triangular with nonzero diagonal r−λ, hence invertible.

In a proposed relation among bases, apply all other roots' factors to isolate the chosen r. Other terms vanish; the surviving polynomial has undergone only invertible maps and must vanish at every nonnegative integer, hence is the zero polynomial. Repeat for each root to prove independence.

General proof for higher-order solutions (continued)

The kernel of E^{m_0} is precisely sequences supported at indices 0,…,m_0-1. Each $E-\lambda$ for nonzero λ is triangular on this space with nonzero diagonal $-\lambda$. The same isolation argument handles the zero-root prefix bases. All k bases are independent and hence span the full solution space. Factor P over the complex numbers; real coefficients permit conjugate pairing. Roots $\rho e^{\pm i\theta}$ give real solutions $\rho^n(A\cos n\theta + B\sin n\theta)$, with real A,B fitted from real data. Repeated roots require additional independent polynomial factors.

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Nonhomogeneous recurrences and resonance

Linearity gives $a_n=a_n^{(h)}+a_n^{(p)}$. First solve the homogeneous equation, choose a particular trial matching the forcing type, substitute and compare coefficients, then add the homogeneous solution and fit initial data to the total. If its base r is a root of multiplicity m, multiply the trial by n^m.

Forcing	Trial before resonance
Constant	Constant
Polynomial degree d	General polynomial degree d
Cr^n	Ar^n
Degree-d polynomial times r^n	General degree-d polynomial times r^n
A cos(nθ)+B sin(nθ)	Both sine and cosine at that frequency

A corresponding root collision multiplies the whole trial by n^m, including conjugate collisions for sinusoidal forcing.

Nonhomogeneous recurrences and resonance (continued)

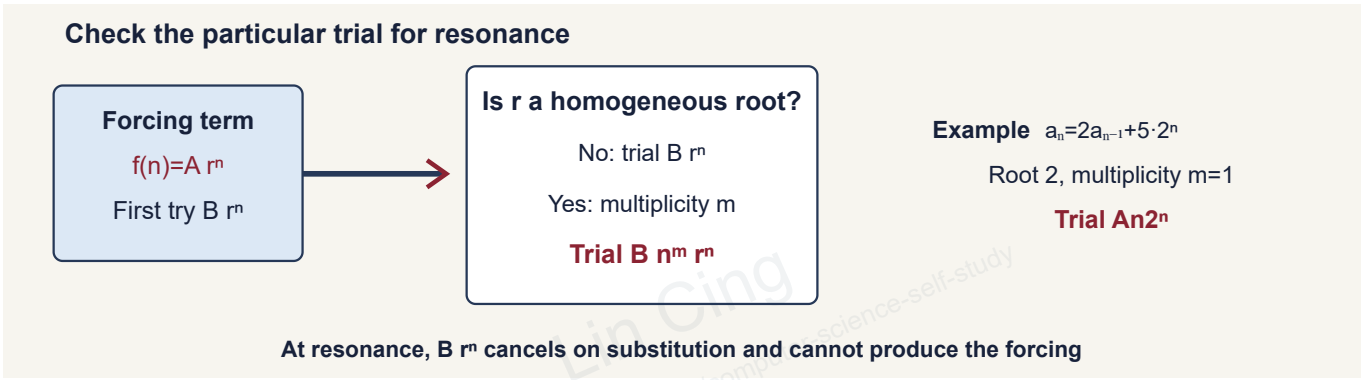
For $a_n=3a_{n-1}+4n, a_0=2$, the homogeneous solution is $C3^n$. Trial $An+B$ gives $An+B=(3A+4)n+(3B-3A)$, so $A=-2, B=-3$. Then $a_n=C3^n-2n-3$, and $a_0=2$ gives $C=5$. Thus $a_n=5\cdot 3^n-2n-3$, with terms $2, 10, 38$; $3\cdot 10+8=38$.

In $a_n-a_{n-1}=2^n$, base two does not collide with root one. In $a_n-2a_{n-1}=2^n$ it does, requiring $Cn2^n$. For $a_n=2a_{n-1}+5\cdot 2^n, a_0=3$, substitute $An2^n$:

$$An2^n = A(n-1)2^n + 5\cdot 2^n, \qquad A=5.$$

Thus $a_n=(C+5n)2^n$ and $C=3$. Terms $a_1=16=2\cdot 3+10$ and $a_2=52=2\cdot 16+20$ check the result.

Nonhomogeneous recurrences and resonance (continued)



Nonhomogeneous recurrences and resonance (continued)

Superposition requires linearity; it does not justify adding particular solutions for nonlinear equations. The resonance rule follows from difference operators. For nonzero r of multiplicity s, $(E-r)^s$ lowers polynomial degree s times; other factors preserve degree and are invertible on each fixed-degree space. Taking $Q(n)=n^sH(n)$, $\deg H \le d$, the map Δ^s sends n^{s+j} to degree j with nonzero leading coefficient. This is an invertible triangular map to degree $\le d$ polynomials, proving existence for every degree-d polynomial times r^n forcing. At s=0 no n-factor is needed. Nonzero r, constant coefficients, and characteristic zero are required.

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Generating functions and rationality

Finite-order homogeneous constant-coefficient recurrences give $Q(x)A(x)=P(x)$, $Q(0)\neq 0$, by multiplying by x^n and summing. Thus the OGF is rational. Rational forcing OGFs preserve rationality. Linearity alone, especially with variable coefficients, does not suffice.

Conversely, if $A = P/Q$, $Q = 1 + q_1x + \cdots + q_dx^d$, coefficient comparison past deg P gives $a_n + q_1a_{n-1} + \cdots + q_da_{n-d} = 0$. An eventually constant-coefficient recurrence with finitely many exceptions conversely yields a rational OGF. These are formal statements independent of convergence.

For $a_n + C_1a_{n-1} + C_2a_{n-2} = f_n$, $n \ge 2$, put $F(x) = \sum_{n\ge 2} f_nx^n$. The sums are $A - a_0 - a_1x$, $x(A - a_0)$, x^2A. Therefore

$$(1 + C_1x + C_2x^2)A = a_0 + (a_1 + C_1a_0)x + F(x).$$

The denominator reflects recurrence coefficients; initial values and forcing determine the numerator.

Generating functions and rationality (continued)

For $a_0=1, a_1=4, a_n=5a_{n-1}-6a_{n-2}$, roots 2,3 give $C_1+C_2=1, 2C_1+3C_2=4$, hence $C_1=-1, C_2=2$ and $a_n=-2^n+2\cdot 3^n$. The OGF route gives

$$(A-1-4x)-5x(A-1)+6x^2A=0, \qquad A=\frac{1-x}{(1-2x)(1-3x)}=-\frac{1}{1-2x}+\frac{2}{1-3x}.$$

Both methods retain the same numerator and constants.

For $a_0=1, a_n=3a_{n-1}+2$, $A-1=3xA+2x/(1-x)$, giving $A=(1+x)/((1-x)(1-3x))$. Partial fractions give $a_n=2\cdot 3^n-1$, with 1,5,17,53.

For $h_0=0, h_n=h_{n-1}+1/n$, linearity still holds but

$$(1-x)H=\sum_{n\geq 1}x^n/n=-\log(1-x), \qquad H=-\log(1-x)/(1-x).$$

The initial terms are 0,1,3/2,11/6,25/12. Differentiating the forcing series gives $1/(1-x)$; its zero constant fixes $-\log(1-x)$.

Generating functions and rationality (continued)

To prove nonrationality, suppose H rational. Then $R=(1-x)H=-\log(1-x)$ is rational. At $x=1$ a rational function either has a finite limit or a pole of finite integer order $m\geq 1$. Set $t=-\log(1-x)$. As $x\to 1^-$, $t\to\infty$ but $t/(1-x)^{-m}=te^{-mt}\to 0$ for every fixed m. Logarithmic divergence is neither finite nor any pole order, contradiction. A formal rational representation would have the same Taylor expansion and analytic function, so is excluded too. Both examples require initial corrections; rationality depends on coefficients and forcing, not just linearity.

Nonlinear recurrences

Equations such as $a_{n+1}=a_n+a_n^2$, or recurrences with products and quotients, usually have no fixed linear characteristic polynomial. Seek monotonicity, bounds, fixed points, or a reversible substitution. The change $b_n=\log a_n$ requires $a_n>0$ and converts products into sums. A nonlinear problem may ask for asymptotic bounds or invariants rather than a closed form; specify the target.

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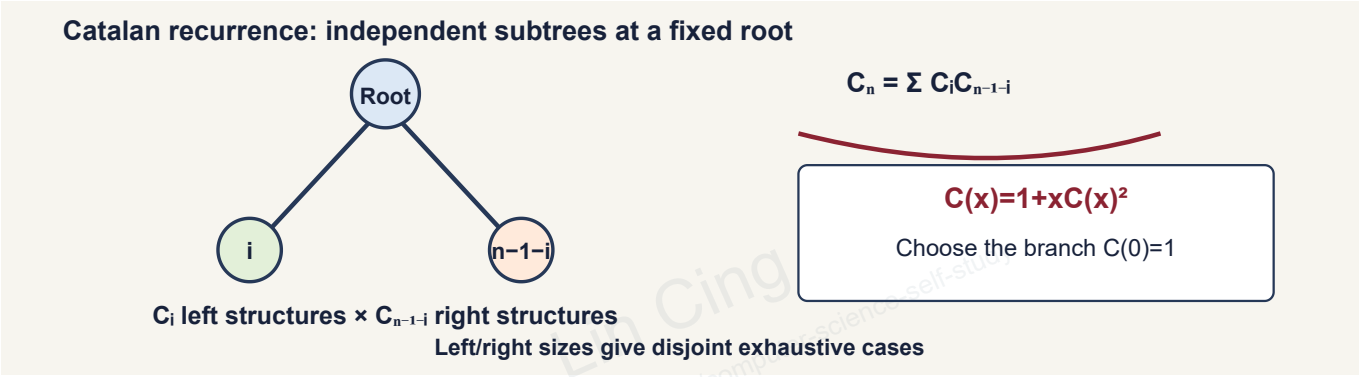
Catalan convolution and a nonlinear closed form

Let C_n count ordered rooted binary trees on n vertices, with $C_0=1$ for the empty tree. A root uniquely joins left and right subtrees of sizes i,n−1−i, giving

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}.$$

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Catalan convolution and a nonlinear closed form (continued)



Summing gives $C-1=xC^2$, so $xC^2-C+1=0$. The branch with constant one is $C = (1 - \sqrt{1-4x})/(2x)$.

Construct the formal square root directly. Let $Y(z) = \sum_{j\geq 0} \binom{1/2}{j} z^j$, with descending-product binomials. Coefficient recurrence $(j+1)c_{j+1}=(1/2-j)c_j$ gives $(1+z)Y'=Y/2, Y(0)=1$. For $F=Y^2$, $(1+z)F'=F, F(0)=1$; coefficient comparison uniquely yields $F=1+z$. Thus Y is the square root with constant one, not the negative branch. Substitute $z=-4x$ and simplify descending products:

Catalan convolution and a nonlinear closed form (continued)

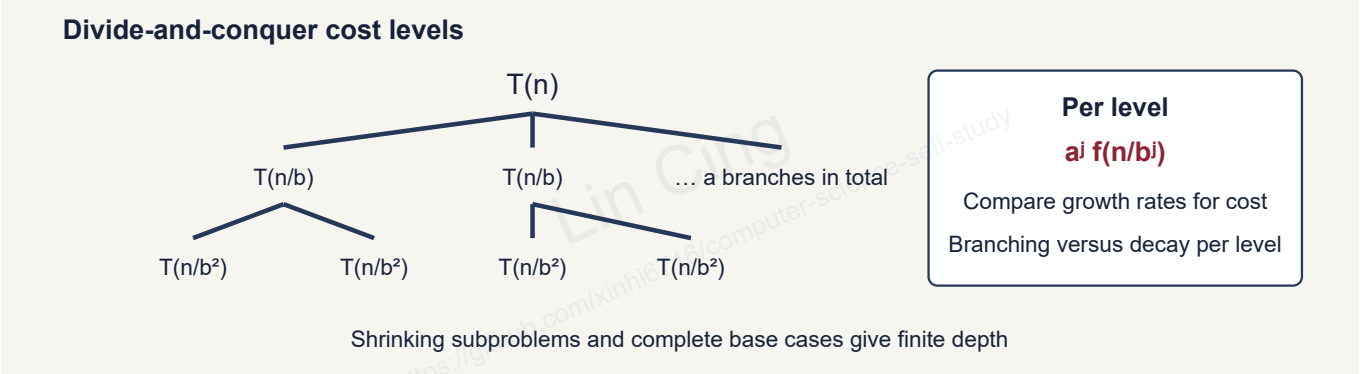
$$\sqrt{1-4x} = 1 - 2\sum_{k\geq 1} \frac{1}{k}\binom{2k-2}{k-1} x^k.$$

Substitution and $n=k-1$ give $C_n = \binom{2n}{n}/(n+1)$, with 1,1,2,5,14,42. At n=3 the five ordered trees biject with five stack outputs of {1,2,3}. Identifying left/right subtrees changes the object and invalidates the same count.

The nonlinear sequence $a_0=1, a_{n+1}=a_n+a_n^2$ gives 1,2,6,42,… and is not the Catalan recurrence. Seek transformations, convolution, or growth invariants rather than characteristic roots.

Divide-and-conquer recurrences

For $T(n) = aT(\lceil n/b \rceil) + f(n)$, handle exact floors/ceilings and ensure subproblems strictly decrease past the base threshold. A polynomial-cost master theorem compares a with b^d for $aT(n/b)+n^d$, identifying leaves, equal-cost layers, or the root as dominant.



Fix integers $a \geq 1, b \geq 2, c > 0$ and $n = b^k$. For $f(n) = af(n/b) + c, f(1) = c$,

$$f(b^k) = a^k f(1) + c(1 + a + \cdots + a^{k-1}).$$

Divide-and-conquer recurrences (continued)

If $a=1$, this is $c(k+1)=\Theta(\log_b n)$. If $a \geq 2$ the sum is $(a^k-1)/(a-1)$, yielding $\Theta(a^k)=\Theta(n^{\log_b a})$. Generally,

$$f(b^k) = a^k f(1) + \sum_{j=0}^{k-1} a^j h(b^{k-j}).$$

For fixed $d \geq 0, h(n)=\Theta(n^d)>0, f(1)=\Theta(1)>0$, the layer ratio a/b^d gives $\Theta(n^{\log_b a})$ when $d<\log_b a$, $\Theta(n^d \log n)$ at equality, and $\Theta(n^d)$ when $d>\log_b a$. This proof is on powers of b. General integers require floor/ceiling and monotonicity bounds between adjacent powers.

For $T(n)=2T(n/2)+n/\log_2 n$ on $n=2^k \geq 2$, a polynomial-forcing theorem cannot replace $n/\log n$ by n. Level j costs $n/(k-j)$, so

$$T(2^k) = 2^k T(1) + 2^k \sum_{j=0}^{k-1} \frac{1}{k-j} = n(T(1) + H_k) = \Theta(n \log \log n).$$

Use a separate positive $T(1)$; do not divide by $\log_2 1$. At $n=16, T(1)=1$, internal levels cost $4, 16/3, 8, 16$ and leaves cost 16, totaling $148/3 = 16(1+H_4)$. A recurrence that still calls itself at $n=1$ has no termination proof.

Master-theorem examples and limits

For positive constant bases, $2T(n/2)+n$ gives $\Theta(n \log n)$, $4T(n/2)+n$ gives $\Theta(n^2)$, and $2T(n/4)+n$ gives $\Theta(n)$. Recursion-tree layer sums verify each. Nonpolynomial forcing or unsuitable a,b and division conditions require a broader theorem, recursion trees, Akra–Bazzi, or direct induction. A Θ claim requires both upper and lower bounds; $O(n \log n)$ alone does not complete a tight analysis.

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Choosing a recurrence method

Use iteration or integrating factors for first order; characteristic roots for homogeneous constant-coefficient linear equations; homogeneous plus particular solutions with resonance checks for forcing; generating functions for coefficient extraction; transformations, invariants, or bounds for nonlinear equations; and exact size recurrences followed by justified tree/theorem/induction analysis for divide and conquer. Characteristic roots outside their linear constant-coefficient family are a mathematical error, not a stylistic choice.

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Knockout tournaments

With $n=2^m$ entrants, each game eliminates one player. Every player except the champion is eliminated once, so $n-1$ games suffice and are necessary. Equivalently $T(1)=0, T(n)=2T(n/2)+1$ unrolls to $n-1$. One argument counts eliminations, the other solves a recurrence; agreement checks the base and merge-cost meanings.

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Coupled recurrences and simultaneous generating functions

For fixed matrix M and $y_n=My_{n-1}+f_n$, let $G=\sum_{n\ge 0} y_n x^n$, $F=\sum_{n\ge 1} f_n x^n$. Then

$$G-y_0=xMG+F, \qquad G=(I-xM)^{-1}(y_0+F).$$

No invertibility of M or analytic convergence is needed. The formal series $\Sigma x^k M^k$ multiplies $I-xM$ to I by cancellation of every positive-degree coefficient. If F has rational components, the adjugate formula and determinant constant one make G's components rational.

For $a_n=2a_{n-1}+b_{n-1}, b_n=a_{n-1}+b_{n-1}, a_0=1, b_0=0$, use the same old pair in both updates.

n	Old pair	New a_n	New b_n
1	(1,0)	$2\cdot 1+0=2$	$1+0=1$
2	(2,1)	$2\cdot 2+1=5$	$2+1=3$
3	(5,3)	$2\cdot 5+3=13$	$5+3=8$
4	(13,8)	$2\cdot 13+8=34$	$13+8=21$

The OGF system is

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Coupled recurrences and simultaneous generating functions (continued)

$$\begin{pmatrix} 1-2x & -x \\ -x & 1-x \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = \frac{1-x}{1-3x+x^2}, \quad B = \frac{x}{1-3x+x^2}.$$

Both components satisfy $u_n=3u_{n-1}-u_{n-2}$ but have distinct initial values. With $r_\pm = (3 \pm \sqrt{5})/2$, partial fractions give

$$b_n = \frac{r_+^n - r_-^n}{\sqrt{5}}, \qquad a_n = \frac{(r_+ - 1)r_+^n + (1 - r_-)r_-^n}{\sqrt{5}}.$$

At n=0,1 these give (1,0),(2,1); uniqueness of each second-order recurrence then proves agreement for all terms. Elimination must preserve each component's own initial values.

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Boundaries and verification

Check initial values and substitute into the original recurrence. Verify index floors/ceilings, eventual scope of asymptotic bounds, stopping conditions, and the operation model. Exercises and solutions provide further derivations. Closed forms are one objective; lower-order terms can matter to actual cost or memory and cannot always be discarded.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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